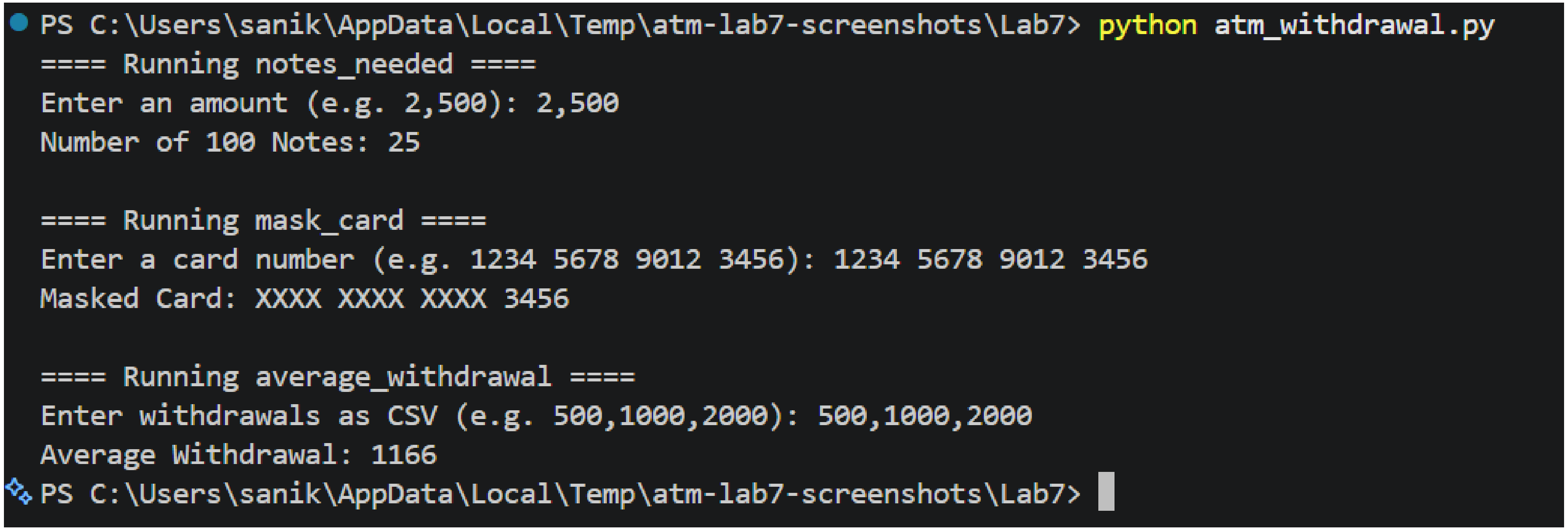
Before the fixes

1. Original ATM application

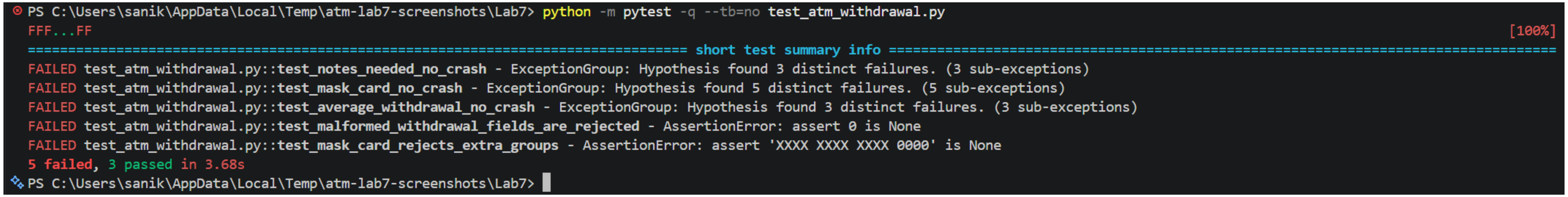


```
PS C:\Users\sanik\AppData\Local\Temp\atm-lab7-screenshots\Lab7> python atm_withdrawal.py
==== Running notes_needed ====
Enter an amount (e.g. 2,500): 2,500
Number of 100 Notes: 25

==== Running mask_card ====
Enter a card number (e.g. 1234 5678 9012 3456): 1234 5678 9012 3456
Masked Card: XXXX XXXX XXXX 3456

==== Running average_withdrawal ====
Enter withdrawals as CSV (e.g. 500,1000,2000): 500,1000,2000
Average Withdrawal: 1166
PS C:\Users\sanik\AppData\Local\Temp\atm-lab7-screenshots\Lab7> 
```

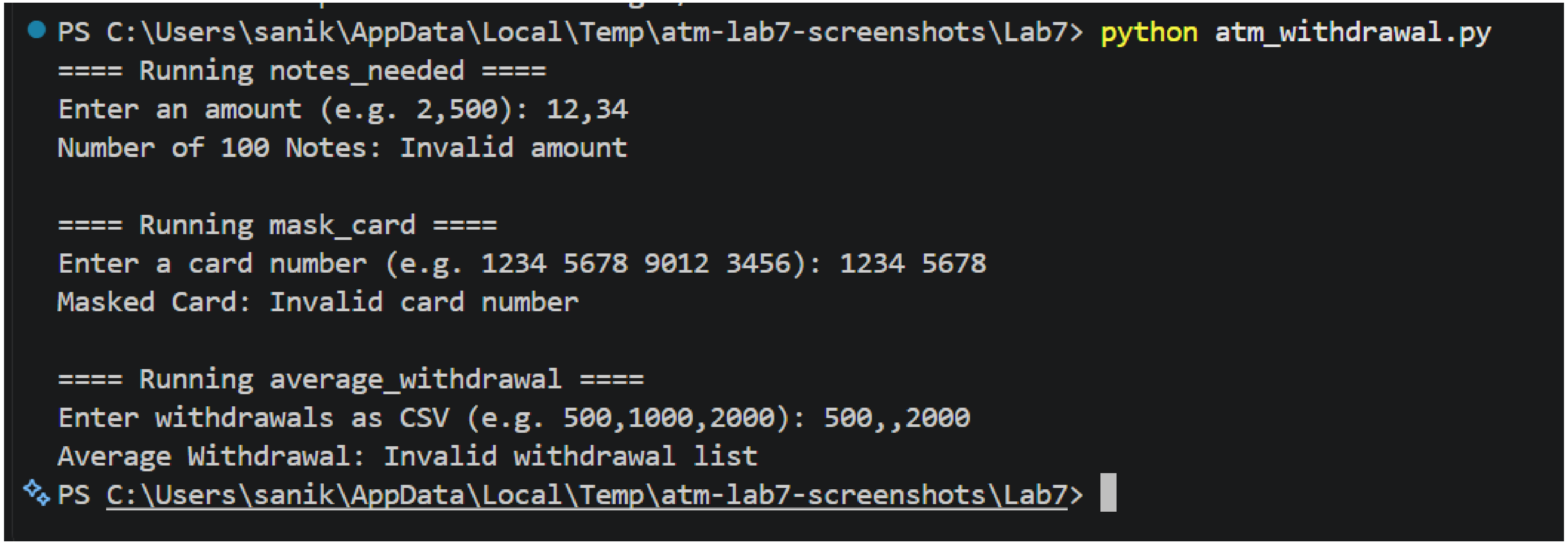
2. Hypothesis tests against the original code



```
PS C:\Users\sanik\AppData\Local\Temp\atm-lab7-screenshots\Lab7> python -m pytest -q --tb=no test_atm_withdrawal.py
FFF...FF                                                                                                                                                                                   [100%]
=================================================================================== short test summary info =====================================================================================
FAILED test_atm_withdrawal.py::test_notes_needed_no_crash - ExceptionGroup: Hypothesis found 3 distinct failures. (3 sub-exceptions)
FAILED test_atm_withdrawal.py::test_mask_card_no_crash - ExceptionGroup: Hypothesis found 5 distinct failures. (5 sub-exceptions)
FAILED test_atm_withdrawal.py::test_average_withdrawal_no_crash - ExceptionGroup: Hypothesis found 3 distinct failures. (3 sub-exceptions)
FAILED test_atm_withdrawal.py::test_malformed_withdrawal_fields_are_rejected - AssertionError: assert 0 is None
FAILED test_atm_withdrawal.py::test_mask_card_rejects_extra_groups - AssertionError: assert 'XXXX XXXX XXXX 0000' is None
5 failed, 3 passed in 3.68s
PS C:\Users\sanik\AppData\Local\Temp\atm-lab7-screenshots\Lab7> 
```

After the fixes

3. Fixed ATM application with invalid inputs

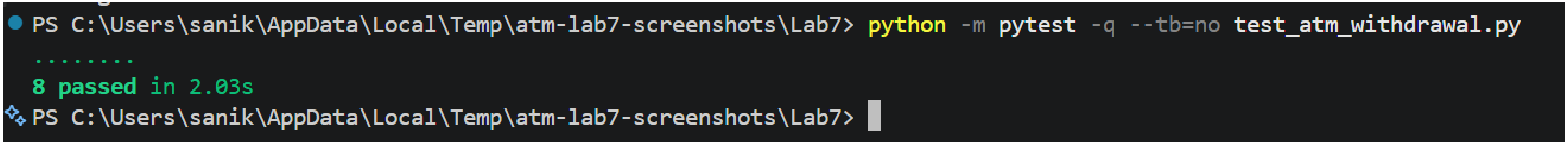


```
PS C:\Users\sanik\AppData\Local\Temp\atm-lab7-screenshots\Lab7> python atm_withdrawal.py
==== Running notes_needed ====
Enter an amount (e.g. 2,500): 12,34
Number of 100 Notes: Invalid amount

==== Running mask_card ====
Enter a card number (e.g. 1234 5678 9012 3456): 1234 5678
Masked Card: Invalid card number

==== Running average_withdrawal ====
Enter withdrawals as CSV (e.g. 500,1000,2000): 500,,2000
Average Withdrawal: Invalid withdrawal list
PS C:\Users\sanik\AppData\Local\Temp\atm-lab7-screenshots\Lab7>
```

4. Hypothesis tests against the fixed code



```
PS C:\Users\sanik\AppData\Local\Temp\atm-lab7-screenshots\Lab7> python -m pytest -q --tb=no test_atm_withdrawal.py
........
8 passed in 2.03s
PS C:\Users\sanik\AppData\Local\Temp\atm-lab7-screenshots\Lab7>
```